



Department of Information Technology
Academic Year 2023 - 2024 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3352 & Foundations of Data Science

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Introduction
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** Think Pair Share
- **Justification:**

In unit II, Describing Data with Averages is an important topic needed for every data scientist because the median, mode, standard deviation are important for describing data. This activity helps the students to know the concepts and implementation methods.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The topic Describing data with averages taught in the previous days.
 - The time allotted for the topic is 20 Minutes.
 - The students were asked to form a team with two pairs. The total teams are 29, the questions are allotted to each team as shown in figure 1,2.
 - Asked the student to discuss with their pairs and find the answer for their corresponding questions. The time allotted is 10 minutes.
 - The students were asked to present their solution as shown in figure 3.
 - First Shanmugapriya and Harshini presented their median and mode for ungrouped data. Next SriSarukumar and Dheenathayalan presented median for Grouped data.
 - Next Vijayalaksmi and Sobana team Dinesh and Suhail Ahmed team presented standard deviation for grouped data as shown in the Figure 4, in the same way all the teams were presented.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	2	1	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	3	2	2	2	1	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of data science for describing data with averages	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow for developing data science applications.	Able to analyze the data and represent the data with valid conclusion	The students able to function as a team in analyzing and describing data.	Intrpret statistical description of data and prepare report	The students able to recognize the need for statistical techniques for describing data.	The students able to exhibit the knowledge in statistics	The students can Exhibit knowledge of data science in the field of IoT, Cloud Computing to develop IT solutions	The students able to exhibit knowledge in statistical techniques of data to develop a quality product.

- **Images / Screenshot of the practice:**



Figure 1: Thin Pair Share

Questions:

- Calculate the standard deviation and mean of a dataset with values: [5, 8, 12, 15, 20].
 - Find the mode of the following numbers: 5, 8, 6, 9, 8, 3, 5, 2, 8.
- Calculate the median for the given grouped data: [10-20]: 5, [20-30]: 12, [30-40]: 8, [40-50]: 6.
 - Calculate the mode of the following test scores: 88, 92, 78, 92, 85, 90, 88, 93.
- In a survey, participants' ages are recorded. Find the median age from the following ages: 29, 31, 35, 40, 41, 45, 49.
 - Find the standard deviation for a dataset with values: [10, 10, 10, 10, 10].
- Find the median for the following dataset: [0-10]: 15, [10-20]: 25, [20-30]: 10.
 - Calculate the mode of the following test scores: 88, 92, 78, 92, 85, 90, 88, 93.
- Determine the standard deviation for a dataset with values: [2, 4, 6, 8, 10].
 - In a survey, participants' ages are recorded. Find the median age, mean from the following ages: 29, 31, 35, 40, 41, 45, 49.
- What is the mode of the shoe sizes for a group of 20 people: 8, 9, 10, 9, 7, 8, 9, 10?
 - Determine the median for grouped data: [0-5]: 8, [5-10]: 12, [10-15]: 20, [15-20]: 15.
- Calculate the standard deviation of a sample with values: [12, 14, 18, 20, 22].
 - Calculate the mean, median for the following dataset of annual incomes: \$45,000, \$50,000, \$55,000, \$60,000, \$75,000, \$80,000, \$100,000.
- Find the median for grouped data: [0-100]: 7, [100-200]: 18, [200-300]: 11.
 - Calculate the mode of the temperatures recorded for a week: 78°F, 82°F, 78°F, 77°F, 83°F, 79°F, 78°F.

9. a) . Determine the median for the grouped data: [0-25]: 3, [25-50]: 9, [50-75]: 16.
b) . Find the mode of the following series: 12, 18, 24, 12, 30, 18, 36, 24.
10. a) Find the standard deviation for a dataset with values: [3, 7, 11, 15, 19].b) .
Find the median height of a group of individuals with the heights (in inches):
62, 64, 66, 68, 70, 72, 74.
11. a) A data set represents the number of hours worked by employees in a week:
32, 40, 42, 45, 50, 55. Calculate the median.
b) Calculate the population standard deviation for a dataset with values:
[8, 16, 24, 32, 40].
12. a) Determine the median for the grouped data: [0-25]: 3, [25-50]: 9, [50-75]: 16.
b) Determine the mode of the following list of book prices:
\$12.50, \$15.75, \$12.50, \$18.25, \$20.50, \$15.75.
13. a) Find the median height of a group of individuals with the heights (in inches):
62, 64, 66, 68, 70, 72, 74.
b) Determine the standard deviation for a dataset with values: [5, 12, 19, 26, 33].
14. a) A data set represents the number of hours worked by employees in a week:
32, 40, 42, 45, 50, 55. Calculate the median.
b) Calculate the standard deviation of a dataset with values:
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10].
15. a) . Calculate the median for the dataset grouped as [0-5]:
12, [5-10]: 14, [10-15]: 8, [15-20]: 11.
b) Calculate the mode of the number of siblings for a group of 50 students:
1, 2, 2, 0, 1, 3, 1, 0.
16. a) Find the standard deviation for a dataset with values:
[0, 5, 10, 15, 20, 25, 30, 35, 40, 45].
b) From a set of daily website visit counts: 200, 250, 300, 350, 400, 450, 600, find the median.
17. a) . Find the median for the grouped data: [0-30]: 20, [30-60]: 15, [60-90]: 10.
b) Find the mode of the ages of a group of people: 35, 27, 41, 35, 29, 29, 41, 35, 38.
18. a) . Determine the median for the dataset grouped as [0-40]: 6, [40-80]: 12, [80-120]: 18.
b) Determine the mode of the number of daily website visitors over a week:
100, 150, 125, 150, 100, 175, 200.
19. a) Calculate the standard deviation of a sample with values: [22, 24, 25, 28, 30, 32].
b) Calculate the mean, median for a dataset representing the response times (in seconds) for a web application: 3.2, 4.5, 5.1, 5.8, 6.4, 7.0.
20. a) . Calculate the mode of the monthly rainfall measurements (in inches) for a year:
2.5, 3.0, 1.5, 2.5, 1.0, 3.5, 2.5, 1.5, 3.0.
b) Determine the standard deviation for a dataset with values: [2, 2, 2, 8, 8, 8].

21. a) Calculate the median for the following grouped data:
[0-10]: 7, [10-20]: 14, [20-30]: 5.
b) Find the mode of the following numbers: 4, 5, 6, 6, 8, 8, 8, 10, 11, 12.
22. a) . Calculate the population standard deviation for a dataset with values: [3, 3, 3, 3, 3].
b) . Calculate the median for a dataset of product prices (in dollars):
\$15, \$20, \$25, \$30, \$35, \$40, \$50.
23. a) Find the median for the grouped data: [0-50]: 8, [50-100]: 10, [100-150]: 12.
b) Determine the mode of the following test scores:
88, 92, 78, 92, 85, 90, 88, 93, 88, 85.
24. a) Determine the standard deviation for a dataset with values: [10, 20, 30, 40, 50].
b) What is the mode of the shoe sizes for a group of 30 people: 8, 9, 10, 9, 7, 8, 9, 10, 11, 10?
25. a) Determine the mean, median for grouped data: [0-20]: 5, [20-40]: 8, [40-60]: 10, [60-80]: 7.
b) Calculate the mode of the temperatures recorded for a month:
78°F, 82°F, 78°F, 77°F, 83°F, 79°F, 78°F, 85°F.
26. a) Determine the median of a dataset representing the sizes (in square feet) of houses in a neighborhood: 1500, 1600, 1700, 1800, 1900, 2000.
b) Find the standard deviation for a dataset with values: [1, 1, 2, 2, 3, 3, 4, 4, 5, 5].
27. a) Find the median of a dataset representing the weights (in pounds) of a group of individuals: 140, 155, 165, 175, 185, 200.
b) Calculate the standard deviation of a sample with values: [17, 19, 21, 24, 26, 28].
28. a) Find the mode of the following series: 12, 18, 24, 12, 30, 18, 36, 24, 18, 36.
b) Calculate the median for the dataset grouped as [0-25]: 15, [25-50]: 20, [50-75]: 10.
29. a) Determine the median for the grouped data:
[0-15]: 12, [15-30]: 8, [30-45]: 10, [45-60]: 14.
b) Determine the mode of the following list of book prices:
\$12.50, \$15.75, \$12.50, \$18.25, \$20.50, \$15.75, \$12.50, \$18.25.

Figure 2: Questions

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a) Determine the median of a dataset representing the sizes (in square feet) of houses in a neighbourhood 1500, 1600, 1700, 1800, 1900, 2000

Formula for even

$$= \frac{1}{2} \left\{ \left(n/2 \right)^{\text{th}} \text{ term} + \left(n/2 + 1 \right)^{\text{th}} \text{ term} \right\}$$

$$= \frac{1}{2} \left\{ \left(6/2 \right)^{\text{th}} \text{ term} + \left(6/2 + 1 \right)^{\text{th}} \text{ term} \right\}$$

$$= \frac{1}{2} \left\{ 3^{\text{th}} \text{ term} + 4^{\text{th}} \text{ term} \right\}$$

$$= \frac{1}{2} \{ 1700 + 1800 \}$$

$$= \frac{1}{2} \{ 3500 \}$$

Median = 1750.

b) Find the standard deviation for a dataset with values [1, 1, 0, 2, 3, 3, 4, 4, 5, 5]

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{n}}$$

$$\sum x = 1+1+0+2+3+3+4+4+5+5$$

$$\sum x = 30$$

$$\mu = \frac{\sum x}{N} = \frac{30}{10} = 3$$

$$x - \mu = -2, -2, -1, -1, 0, 0, 1, 1, 2, 2$$

$$\sum (x - \mu)^2 = 4+4+1+1+0+0+1+1+2+2$$

$$\sum (x - \mu)^2 = 16$$

$$\sigma = \sqrt{\frac{16}{10}} = \sqrt{1.6}$$

$$\sigma = 1.2649$$

Figure 3 Solution given by Aarthi G and Abinaya S



Figure 4 Presentation by Dinesh and Suhail Ahmed Team Vijayalakshmi and Sobana Team

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students said that they were able to understand the concepts of describing data with averages
- The students said that the activity help to practice the concept such as Mean, Median, Mode and Standard Deviation for various types of data.
- The students were shared their ideas for finding median, mode and standard deviation

- It makes the students to improve presentation skills.
- ❖ ***Benefit of the practice:***
 - Students can able to recall the methods for finding Mean, Median, Mode, Standard Deviation and it improves the problem-solving skills.
 - The students can share their ideas with their pairs.
 - It helps the slow learners to understand the concepts.
- ❖ ***Challenges faced in implementation:***
 - The Activity planned for 20 minutes but it takes 30 minutes to complete the activity.
 - Some students hesitate to present their solution.

References:

- ❖ <https://ohioline.osu.edu/>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

HOD